PYTHON 90 DAYS CHALLENGE

DAY 16 — COLLECTIONS MODULE IN PYTHON

9.5 Collections Module in Python

The **collections** module provides special data structures that are more powerful or convenient for common tasks than using basic lists, tuples, dictionaries, and sets alone.

Why is it useful?

• It provides ready-made advanced data structures.

• It helps with counting, grouping, ordering, fast queue operations, and combining dictionaries.

Tool	Main use
Counter	Counting the frequency of items
defaultdict	Dictionary that supplies default values for missing keys
OrderedDict	Dictionary with extra order-related operations
deque	Fast queue and stack operations at both ends
ChainMap	Combines multiple dictionaries into one view

9.5.1 Counter

Counter is used to count how many times each value appears. It returns a dictionary-like object where each item is a key and its frequency is the corresponding value.

How it works

• It checks the items in the input data.

• It counts repeated items.

• It returns item-and-count pairs.

• It is commonly used for frequency counting, such as counting letters or repeated items in a list.

Common Counter methods

Method	Meaning
most_common()	Returns items ordered by frequency.
most_common(n)	Returns the top n most frequent items.
elements()	Returns items repeated according to their counts.
update()	Adds counts from more data.
subtract()	Subtracts counts using more data.

9.5.2 defaultdict

defaultdict is a dictionary that provides a default value when a key does not exist. A normal dictionary raises a **KeyError** for a missing key, while defaultdict can create a default value automatically.

Key points

• It helps avoid manual checks for missing keys.

• It automatically creates a default value for a missing key.

• Common default types include int, list, set, and str.

Default type	Default value	Common use
int	0	Counting
list	Empty list: []	Grouping values
set	Empty set	Grouping unique values
str	Empty string: ""	Empty text

Normal dictionary vs defaultdict

Point	Normal dictionary	defaultdict
Missing key	Raises KeyError	Creates a default value
Best for	Simple key-value data	Grouping and counting
Manual checking	Often needed	Usually not needed

9.5.3 OrderedDict

OrderedDict stores key-value pairs in order and provides extra methods for changing that order. Modern Python dictionaries also preserve insertion order, but OrderedDict remains useful when order-specific operations are needed.

• It can move an item to the beginning or end.

• It can remove items from either end.

• It is useful for order-sensitive dictionary logic.

Method	Meaning
move_to_end(key)	Moves the selected key to the end.
move_to_end(key, last=False)	Moves the selected key to the beginning.
popitem()	Removes the last item.
popitem(last=False)	Removes the first item.

9.5.4 deque

deque means **double-ended queue**. It allows fast adding and removing of items from both the left and right ends.

• It can be used as a queue or a stack.

• Lists are slower when removing items from the beginning.

• It is useful for queues, recent history, undo operations, and sliding windows.

Common deque methods

Method	Meaning
append()	Adds an item at the right end.
appendleft()	Adds an item at the left end.
pop()	Removes an item from the right end.
popleft()	Removes an item from the left end.
extend()	Adds multiple items at the right end.
extendleft()	Adds multiple items at the left end.
rotate(n)	Rotates the items by n positions.
clear()	Removes all items.

List vs deque

Operation	List	deque
Add at end	Fast	Fast
Remove from end	Fast	Fast
Add at beginning	Slower	Fast
Remove from beginning	Slower	Fast
Best for	General storage	Queue operations

9.5.5 ChainMap

ChainMap combines multiple dictionaries into one view without merging them permanently.

• It groups multiple dictionaries together as a combined view.

• When searching for a key, Python checks dictionaries from left to right.

• If the same key exists in multiple dictionaries, the value from the first dictionary is used.

• It is useful for layered settings, such as user preferences overriding default settings.

Useful ChainMap features

Feature	Meaning
maps	Shows the dictionaries included in the ChainMap.
new_child()	Adds a new dictionary at the front of the view.
parents	Returns a view without the first dictionary.